Adelges tsugae (Hemlock Woolly Adelgid)

The hemlock woolly adelgid (*Adelges tsugae*) is a serious threat to our state tree, the eastern hemlock, in Pennsylvania and across the United States.

This non-native invasive insect has caused significant hemlock defoliation and mortality in Pennsylvania forests.

Adelgids are a small family of insects closely related to aphids and feed on plant sap.

Feeding from adelgids interferes with the tree's use of nutrients, and in the case of hemlock, causes:

- Needle drop
- Branch dieback
- Tree mortality

The egg sacs of these insects look like the tips of cotton swabs clinging to the undersides of the current year's hemlock twigs.

The hemlock woolly adelgid was accidentally introduced to Virginia from Japan in the 1950s, and by the late 1960s was reported in southeastern Pennsylvania.

By the end of 2023, the pest was found in all 67 Pennsylvania counties.

Management of Hemlock Woolly Adelgid Infestations in Pennsylvania Forests

In an effort to forestall the impact of hemlock woolly adelgid, DCNR's Bureau of Forestry has developed the Eastern Hemlock Conservation Plan, and has been treating high-value hemlocks in state parks and forests since 2004.

To manage hemlock woolly adelgid in Pennsylvania's forests, the DCNR Bureau of Forestry uses integrated pest management principles that rely on surveying and monitoring of the insect and its hemlock host, including the following methods:

- Biological control
- Insecticides
- Silvicultural
- Tree breeding for host resistance

The choice of method will vary depending on the site and other circumstances of each situation.

Biological Control

Biocontrol of hemlock woolly adelgid is used in forest situations, on vigorous trees with accessible lower branches that are infested with hemlock woolly adelgid.

The Bureau of Forestry has been introducing several different predatory beetles that feed solely on hemlock woolly adelgid since 1999, thereby reducing the impact of the insect.

Insecticides

Research has shown that adelgid-killing systemic insecticides injected into tree boles or applied to the ground as a soil drench or soil injection will kill hemlock woolly adelgid, and temporarily prevent the establishment of new infestations for up to five to seven years.

This has prompted DCNR to institute a suppression program on state forests and parks. This type of control is restricted to large, high-value (ecological, historical, or aesthetic) trees and hemlock sites.

Insecticides can be used as a "stop-gap" measure to prevent tree mortality and give biological control time to take effect.

Silvicultural Control

DCNR is attempting to restore areas that have been impacted by the hemlock woolly adelgid.

This sometimes involves replanting with native species, such as the eastern white pine, that are similar ecologically, but are not affected by the hemlock woolly adelgid.

Tree Breeding for Host Reistance

DCNR is working with the USDA Forest Service, other states, and several universities to find and develop hemlock trees that are resistant to the hemlock woolly adelgid.

A tree breeding program will be developed to produce seedlings that can be used to restore hemlock sites.

Tips for Homeowners With Hemlock Woolly Adelgid Infestations

What can be done depends on the value of the trees you wish to protect. Individual ornamental trees, small trees, or even larger hemlocks in a landscape environment can be treated with insecticides.

There are several spray materials registered for application to hemlocks by ground-spraying equipment and by injection techniques; however, please note that these can be expensive and results can vary.

Forest landowners with dozens or hundreds of infested hemlocks have a few options:

- Use systemic insecticides
- Harvest and replant
- Wait and see

Most Pennsylvania landowners are watching and hoping that their hemlocks survive.

Some are selling their commercial hemlock timber, knowing that dead hemlock degrades very rapidly.

Other Threats to Hemlock Trees

Elongate hemlock scale (*Fiorinia externa*) is another non-native insect pest of eastern hemlock in the U.S. Originally from Japan, it was first discovered in the United States in Long Island, New York in 1908.

It occurs in many states in the eastern U.S., including Pennsylvania.

Like hemlock woolly adelgid, elongate hemlock scale is a sap feeder, which also can cause tree decline and mortality.

DCNR has been releasing a biological control agent for elongate hemlock scale, as well as using a systemic insecticide to reduce populations on high value hemlocks sites in our state forests and state parks.

Additional Resources

- DCNR Eastern Hemlock Conservation Plan (PDF)
- 2014–2018 Hemlock Woolly Adelgid National Initiative Strategic Plan (PDF)
- DCNR Forest Health Fact Sheet: Hemlock Woolly Adelgid (PDF)
- Map of Hemlock Woolly Adelgid Infested Counties (PDF)
- USDA Biology and Control of Hemlock Woolly Adelgid (PDF)
- Penn State Fact Sheet on Elongate Hemlock Scale (PDF)